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36 (SEM-II) IT 121

2026

PROGRAMMING FOR PROBLEM SOLVING

Paper : IT-121

Full Marks : 100

Time : Three hours

The figures in the margin indicate full marks for the questions.

1. Define the following with examples :
✓ 2×5=10

- (i) Variable
- (ii) Pointer
- (iii) Array
- (iv) Structure
- (v) Function

2. ✓ Explain the difference between while and do... while loops with suitable examples.

10

3. ✓ Write a C program to find the factorial of a number using recursion. 10

4. ✓ Write a C program to swap two numbers using a function and pointers. 10

5. Write a C program to merge two sorted arrays into a single sorted array. 10

6. ✓ Analyse the following code and write the output. Explain step by step. 10

```
#include <stdio.h>
int main()
{
    int a = 5, b = 2;
    printf("%d\n", a++ * ++b);
    return 0;
}
```

7. Write a C program that count the number of vowels in a string. Your program should count the number of vowels regardless of case. ✓ 10

8. Design and implement a C program that simulates a simple student record system using structures. The program should allow : ✓ 15

(a) Adding new student records.

- (b) Displaying all records.
- (c) Searching by roll number.

9. Answer the following questions : $5 \times 3 = 15$

- (a) Compare static memory allocation and dynamic memory allocation in C. Which is more efficient in different scenarios? Justify with examples.
- (b) Evaluate the trade-offs between using arrays and pointers for string manipulation in C. Which approach is more efficient and why?
- (c) Write a C program to display the content of a file. The file name should be provided as command line argument.

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FILE *fp;
